

Envelope (wherever “...” is written)

$$\frac{g^4}{2N_c} \int_0^\tau d\tau' \int \frac{d^2 k_\perp}{(2\pi^2)} \int \frac{d^2 l_\perp}{(2\pi^2)} \int d^2 u_\perp \int d^2 v_\perp e^{i(x_\perp - u_\perp)k_\perp} e^{i(y_\perp - v_\perp)l_\perp} \\ \dots \\ f^{bca} f^{dea} \langle \alpha_{1b}^i(u_\perp) \alpha_{1d}^m(v_\perp) \rangle \langle \alpha_{2c}^j(u_\perp) \alpha_{2e}^n(v_\perp) \rangle$$

G and h functions in the GBW model

$$G_{\text{GBW}}(r^2) = \frac{Q_s^2}{g^2 N_c} \frac{1 - \exp\left\{-\frac{Q_s^2 r^2}{4}\right\}}{Q_s^2 r^2 / 4}$$

$$h_{\text{GBW}}(r^2) = 0$$

$$\frac{dE}{dx}$$

$$\begin{aligned}\frac{dE}{dx} &= -\frac{v}{T} \frac{v^\alpha v^\beta}{v^2} X^{\alpha\beta} \\ &= \dots - \frac{v}{T} \frac{1}{v^2} \left[(v^1)^2 J_0^K J_0^L \delta^{ij} \delta^{mn} - (v_\perp \times \hat{k})(v_\perp \times \hat{l}) J_1^K J_1^L \varepsilon^{ij} \varepsilon^{mn} \right] \dots\end{aligned}$$

$$\begin{aligned}\left(\frac{dE}{dx}\right)_{\text{GBW/MV}} &= -\frac{1}{vT} \frac{g^4}{4} (N_c^2 - 1) \int_0^\tau d\tau' \int_0^{2\pi} \frac{d\phi}{2\pi} \int_0^{2\pi} \frac{d\varphi}{2\pi} \\ &\quad \left[(v^1)^2 \left(G_1(r) G_2(r) + h_1(r) h_2(r) \right) \right. \\ &\quad \left. + (v_\perp)^2 \sin \phi \sin \varphi \left(G_1(r) G_2(r) - h_1(r) h_2(r) \right) \right]\end{aligned}$$

$$\text{with } r^2 = (\tau - \tau')^2 + \tau^2 - 2(\tau - \tau')\tau \cos(\varphi - \phi) + (v_\perp)^2 \tau'^2 + 2|v_\perp| \tau' [(\tau - \tau') \cos \varphi - \tau \cos \phi]$$

$$\begin{aligned}\left(\frac{dE}{dx}\right)_{\text{GBW}} &= -\frac{1}{vT} \frac{g^4}{4} (N_c^2 - 1) \int_0^\tau d\tau' \int_0^{2\pi} \frac{d\phi}{2\pi} \int_0^{2\pi} \frac{d\varphi}{2\pi} \\ &\quad G_{\text{GBW}}^1(r^2) G_{\text{GBW}}^2(r^2) \left((v^1)^2 + (v_\perp)^2 \sin \phi \sin \varphi \right)\end{aligned}$$

$$\text{Pure transverse movement: } v^1 = 0, \quad v_\perp^2 = v^2$$

$$\left(\frac{dE}{dx}\right)_\perp = \dots \frac{1}{Tv} (v_\perp \times \hat{k})(v_\perp \times \hat{l}) J_1^K J_1^L \varepsilon^{ij} \varepsilon^{mn} \dots$$

$$\text{Pure longitudinal movement: } v^1 = v, \quad v_\perp = 0$$

$$\left(\frac{dE}{dx}\right)_\parallel = \dots - \frac{v}{T} J_0^K J_0^L \delta^{ij} \delta^{mn} \dots$$

$$\hat{q}$$

$$\begin{aligned}\hat{q} &= \frac{2}{v} \left(\delta^{\alpha\beta} - \frac{v^\alpha v^\beta}{v^2} \right) X^{\alpha\beta} \\ &= \dots \frac{2}{v} \left[\frac{v_\perp^2}{v^2} (\delta^{ij} \delta^{mn} + v^2 \varepsilon^{ij} \varepsilon^{mn}) J_0^K J_0^L \right. \\ &\quad + i(v_\perp \cdot \hat{l})(\delta^{ij} \delta^{mn} - \varepsilon^{ij} \varepsilon^{mn}) J_0^K J_1^L \\ &\quad + i(v_\perp \cdot \hat{k})(\delta^{ij} \delta^{mn} - \varepsilon^{ij} \varepsilon^{mn}) J_1^K J_0^L \\ &\quad \left. - \left((v^1)^2 (\hat{k} \cdot \hat{l}) + (v_\perp \cdot \hat{k})(v_\perp \cdot \hat{l}) \right) \left(\delta^{ij} \delta^{mn} + \frac{1}{v^2} \varepsilon^{ij} \varepsilon^{mn} \right) J_1^K J_1^L \right] \dots\end{aligned}$$

$$\begin{aligned}\hat{q}_{\text{GBW/MV}} &= \frac{g^4}{2} (N_c^2 - 1) \int_0^\tau d\tau' \int_0^{2\pi} \frac{d\phi}{2\pi} \int_0^{2\pi} \frac{d\varphi}{2\pi} \\ &\quad \frac{1}{v} \left[\frac{(v_\perp)^2}{v^2} \left(G_1 G_2 (1 + v^2) + h_1 h_2 (1 - v^2) \right) \right. \\ &\quad + \left(\frac{(v^1)^2}{v^2} \cos(\phi - \varphi) + \frac{(v_\perp)^2}{v^2} \cos \phi \cos \varphi \right) (G_1 G_2 (1 + v^2) + h_1 h_2 (v^2 - 1)) \\ &\quad \left. - 2|v_\perp| h_1 h_2 (\cos \phi + \cos \varphi) \right] \\ &= \frac{g^4}{2} (N_c^2 - 1) \int_0^\tau d\tau' \int_0^{2\pi} \frac{d\phi}{2\pi} \int_0^{2\pi} \frac{d\varphi}{2\pi} \\ &\quad \frac{1}{v} \left[\frac{1 + v^2}{v^2} G_1 G_2 \left((v^1)^2 \cos(\phi - \varphi) + (v_\perp)^2 (\cos \phi \cos \varphi + 1) \right) \right. \\ &\quad - \frac{1 - v^2}{v^2} h_1 h_2 \left((v^1)^2 \cos(\phi - \varphi) + (v_\perp)^2 (\cos \phi \cos \varphi - 1) \right) \\ &\quad \left. - 2|v_\perp| h_1 h_2 (\cos \phi + \cos \varphi) \right]\end{aligned}$$

$$\text{with } r^2 = (\tau - \tau')^2 + \tau^2 - 2(\tau - \tau')\tau \cos(\varphi - \phi) + (v_\perp)^2 \tau'^2 + 2|v_\perp| \tau' [(\tau - \tau') \cos \varphi - \tau \cos \phi]$$

$$\hat{q}_{\text{GBW}} = \frac{g^4}{2} (N_c^2 - 1) \frac{1 + v^2}{v^3} \int_0^\tau d\tau' \int_0^{2\pi} \frac{d\phi}{2\pi} \int_0^{2\pi} \frac{d\varphi}{2\pi} \quad (1)$$

$$G_{\text{GBW}}^1(r^2) G_{\text{GBW}}^2(r^2) \left((v^1)^2 \cos(\phi - \varphi) + (v_\perp)^2 (\cos \phi \cos \varphi + 1) \right)$$

Pure transverse movement: $v^1 = 0$, $v_\perp^2 = v^2$

$$\begin{aligned} \hat{q}_\perp = & \dots \frac{2}{v} \left[(\delta^{ij} \delta^{mn} + v^2 \varepsilon^{ij} \varepsilon^{mn}) J_0^K J_0^L \right. \\ & + i(v_\perp \cdot \hat{l})(\delta^{ij} \delta^{mn} - \varepsilon^{ij} \varepsilon^{mn}) J_0^K J_1^L \\ & + i(v_\perp \cdot \hat{k})(\delta^{ij} \delta^{mn} - \varepsilon^{ij} \varepsilon^{mn}) J_1^K J_0^L \\ & \left. - (v_\perp \cdot \hat{k})(v_\perp \cdot \hat{l}) \left(\delta^{ij} \delta^{mn} + \frac{1}{v^2} \varepsilon^{ij} \varepsilon^{mn} \right) J_1^K J_1^L \right] \dots \end{aligned}$$

Pure longitudinal movement: $v^1 = v$, $v_\perp = 0$

$$\hat{q}_\parallel = \dots - \frac{2}{v} (\hat{k} \cdot \hat{l}) (v^2 \delta^{ij} \delta^{mn} + \varepsilon^{ij} \varepsilon^{mn}) J_1^K J_1^L \dots$$

For Pablo:

$$\begin{aligned}
\frac{dE}{dx} = & \frac{g^4}{2N_c} \int_0^\tau d\tau' \int \frac{d^2 k_\perp}{(2\pi^2)} \int \frac{d^2 l_\perp}{(2\pi^2)} \int d^2 u_\perp \int d^2 v_\perp e^{i(x_\perp - u_\perp)k_\perp} e^{i(y_\perp - v_\perp)l_\perp} \\
& \left(-\frac{v}{T} \frac{1}{v^2} \left[(v^1)^2 J_0^K J_0^L \delta^{ij} \delta^{mn} - (v_\perp \times \hat{k})(v_\perp \times \hat{l}) J_1^K J_1^L \varepsilon^{ij} \varepsilon^{mn} \right] \right. \\
& \left. f^{bca} f^{dea} \langle \alpha_{1b}^i(u_\perp) \alpha_{1d}^m(v_\perp) \rangle \langle \alpha_{2c}^j(u_\perp) \alpha_{2e}^n(v_\perp) \rangle \right) \\
\hat{q} = & \frac{g^4(N_c^2 - 1)}{v} \int_0^t dt' \int_{k_\perp q_\perp} \int_{u_\perp v_\perp} e^{ik_\perp(x-u)_\perp} e^{iq_\perp(y-v)_\perp} W_1^{ik}(u_\perp, v_\perp) W_2^{jl}(u_\perp, v_\perp) \\
& \left[\frac{v_\perp^2}{v^2} (\delta^{ij} \delta^{kl} + v^2 \varepsilon^{ij} \varepsilon^{kl}) J_0(kt) J_0(q(t-t')) \right. \\
& + i \frac{v_\perp \cdot q}{q} (\delta^{ij} \delta^{kl} - \varepsilon^{ij} \varepsilon^{kl}) J_0(kt) J_1(q(t-t')) \\
& + i \frac{v_\perp \cdot k}{k} (\delta^{ij} \delta^{kl} - \varepsilon^{ij} \varepsilon^{kl}) J_1(kt) J_0(q(t-t')) \\
& \left. - \left((v^1)^2 \frac{k \cdot q}{kq} + \frac{(v_\perp \cdot k)(v_\perp \cdot q)}{kq} \right) \left(\delta^{ij} \delta^{kl} + \frac{1}{v^2} \varepsilon^{ij} \varepsilon^{kl} \right) J_1(kt) J_1(q(t-t')) \right]
\end{aligned}$$